

Detecting Spectral Temporal Patterns of Changes from Grasslands to Arable Lands using LUCAS Data

Provided scripts has been created as a part of Ph.D. project that is aiming to map changes from grasslands to arable lands. The main purpose of developed R scripts is to provide users to extract spectral temporal patterns from the LUCAS point database that represent changes from grasslands to arable lands and satellite data from Google Earth Engine.

Installation

In the first place, there is a necessity to install required packages. All necessary packages to run each individual scripts are at the beginning of each script („library“ command). All scripts were tested in R version 4.2.3 in Microsoft Windows environment.

Input Data and their Structure

Sample data are points from LUCAS database that represent changes from grasslands to individual crops. There are two files: „*E20_original_bands.csv*“ and „*NEW_Data_2026.csv*“. The file „*E20_original_bands.csv*“ contains extracted values of original bands of Landsat satellite time series from 2006 to 2016. The second one „*NEW_Data_2026.csv*“ contains values of vegetation indices EVI, NDVI and NDWI vegetation indices. The attribute field „*Change_Vec*“ is type of change from grasslands to individual crop based on code of LUCAS point database.

Processing Chain how to Extract Spectral Temporal Patterns

The first step of the processing chain is to separate and filter original data based on the ID of each LUCAS sample for original bands as well as vegetation indices. The first file „*Data_Frame_LUCAS_GEE_Creation_AND_Filter_ALL_IDS_LANDSAT.R*“ generates individual .csv files named by its original IDs in the working directory. The same thing does the second file „*Data_Frame_LUCAS_GEE_Creation_AND_Filter_ALL_IDS_LANDSAT_NEW_2026.R*“ for the vegetation indices. All original values of spectral bands and vegetation indices are scaled between 0 and 1.

The second step is visualisation of spectral temporal patterns for each LUCAS sample ID for both original Landsat bands and calculated vegetation indices. The first file „*Spectral Curves Plot_ORIGINAL_BANDS.R*“ creates .jpg plots for each LUCAS sample ID based on directory from the first step, where original .csv files are located. Analogically script called „*Spectral Curves Plot_VEGETATION_INDICES.R*“ creates such plots for vegetation indices.

The third step of processing chain implements change detection process based on extracted vegetation indices with the help of *Bfast01* function from *Bfast* algorithm. The change detection process can be applied based on chosen vegetation index. Therefore three R scripts must be used individually in order to detect changes from grasslands to crops correctly. Three R scripts for this purpose are at disposal:

- „BFAST01_Change_Detection_BATCH_LANDSAT_EVI.R“,
- „BFAST01_Change_Detection_BATCH_LANDSAT_NDVI.R“
- „BFAST01_Change_Detection_BATCH_LANDSAT_NDWI.R“

Running each of these scripts individually will extract exact date, when change might occur including its visual chart. Overall scheme for the whole processing chain is visible in the picture below:

